

The development of AgriTrace followed an Agile methodology with iterative development cycles. The project was divided into sprints, each focusing on specific functional modules.

6.1 Development Approach

The Agile Scrum methodology was adopted for its flexibility and iterative nature. The development was organized into the following phases:

1. **Requirement Gathering:** Studied the agricultural waste management ecosystem, identified stakeholder needs (farmers, agents, administrators), and defined functional and non-functional requirements.
2. **System Design:** Designed the system architecture using a component-based approach with Next.js App Router. Created data models for Firestore collections and defined security rules for role-based access control.
3. **Frontend Development:** Built the user interface using React components with TypeScript, styled using Tailwind CSS and Radix UI. Implemented responsive layouts for all device sizes.
4. **Backend Integration:** Configured Firebase services (Firestore, Authentication, Storage), implemented API routes for payment processing, and integrated location and notification services.
5. **Feature Implementation:** Developed core features including waste reporting, listing marketplace, payment integration, carbon credit tracking, and the rewards system.
6. **Testing and Debugging:** Performed unit testing, integration testing, and user acceptance testing. Applied ESLint and TypeScript type-checking for code quality.
7. **Deployment:** Deployed the application to Firebase Hosting with CI/CD pipeline integration.

6.2 Sprint Breakdown

The development was organized into six two-week sprints. The following table summarizes the sprint plan:

6.3 Architecture Pattern

AgriTrace employs a **three-tier architecture**:

- **Presentation Layer:** React components with Next.js App Router, Tailwind CSS styling, and Radix UI primitives for accessible, composable UI elements.